



Andrew Meikle's threshing machine brought to an end the age-old practice of separating grain from stalks and husks with a hand flail.



Swiss naturalist Jean Senebier was one of the first to realize that plants respire. Gases move in and out of leaves through tiny surface pores (stomata), seen here under an electron microscope.

about electrical charges, showing that the force of attraction or repulsion between two charges varies in inverse proportion to the square of the distance between them. **The SI unit of charge, the coulomb, is named after him.**

Also in France in 1785, aviator **Jean Blanchard** (1753–1809) invented the **first folded silk parachute**. Previous parachute designs usually had a heavy wooden frame and failed to work. Blanchard used his parachute successfully to drop a dog in a basket from a balloon high in the air. With American **John Jeffries** (1744–1819), Blanchard made the first airborne crossing of the English Channel, in a gas balloon. In danger of descending too early, the pair shed all of their ballast—

“...LITTLE CAUSES ARE CONSIDERED AS BRINGING ABOUT THE GREATEST CHANGES.”

James Hutton, Scottish geologist, from *Theory of the Earth*, 1795

and most of their clothes—to keep the balloon in the air.

In the same year, Scottish engineer **Andrew Meikle** (1719–1811) invented a **threshing machine**, and English inventor **Edmund Cartwright** (1743–1823) developed a **steam-powered loom**. Both machines dramatically reduced the need for manual labor, causing protests from laid-off workers.

GERMAN-BORN BRITISH ASTRONOMER WILLIAM HERSCHEL

HERSCHEL was the greatest astronomer of his age. He conducted many deep-sky surveys of objects in space with the powerful telescopes he designed himself, and made many thousands of discoveries. On January 11, 1787, he observed that **Uranus**, the planet he had discovered just six years earlier (see 1781–82), **had two moons**, which were named Oberon and Titania, after the fairy king and queen in Shakespeare's *A Midsummer Night's Dream*.

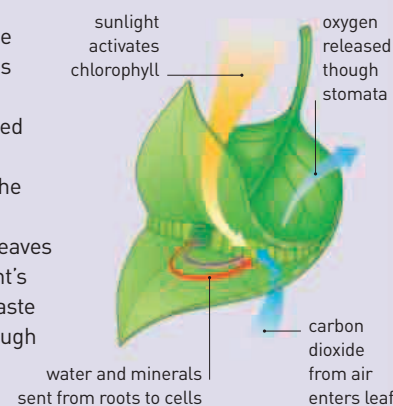
Swiss naturalist **Jean Senebier** (1742–1809) built on the discoveries made about photosynthesis by Jan Ingenhousz in 1779. Senebier showed that when plants are exposed to light, they take up carbon dioxide from the air and release oxygen (see panel, right).

Silicon is the second most abundant element in Earth's crust after oxygen, but it was not identified until 1787. In that year, French chemist **Antoine Lavoisier** understood that sand is an oxide (a chemical combination with oxygen) of a **hitherto unknown element, which he named silicon**.

Also in this year, various inventors were trying to **harness steam power to propel boats**.

PHOTOSYNTHESIS

Plants take carbon dioxide from the air in the process called photosynthesis. Sunlight, which is absorbed by chlorophyll, the green pigment in leaves, fuels the reaction between carbon dioxide and water in the leaves to make glucose, the plant's energy food. Oxygen, a waste byproduct, is exuded through pores (stomata) on the underside of leaves.



American inventor **John Fitch** (1743–98) launched the *Perseverance*, a **boat rowed by steam-driven oars**, on the Delaware River. Scottish engineer **William Symington** (1764–1831) built a **paddle steamer**, which had its first trial on Dalswinton Loch in Scotland. In West Virginia, on the Potomac River, American engineer **James Rumsey** (1743–92) experimented with a boat propelled by a jet of water forced out by a steam pump.

In 1788, French **mathematician Joseph Louis Lagrange** (1736–1813) published his *Mécanique Analytique* (*Analytic Mechanics*), perhaps the greatest work about mechanics (the science of forces and movement) since Isaac Newton's *Principia* (see 1685–89). In this book, Lagrange used his own **new method of calculus** to reduce mechanics to a few basic formulas from which everything else could be calculated.

5 MILES PER HOUR THE SPEED OF SYMINGTON'S PADDLE STEAMER

JAMES HUTTON (1726–97)



After inheriting his father's farm in Berwickshire, Scotland, in 1753, Edinburgh-born James Hutton became fascinated by geology. He studied rock formations and noticed breaks in the sequence, which suggested to him, correctly, that the landscape is shaped in repeated cycles over an immensely long time.

1786 American scientist Benjamin Franklin's 1770 map of the Gulf Stream is published in the US

January 11, 1787 William Herschel discovers Uranus's moons

1787 Jean Senebier makes discoveries about plant photosynthesis

August 22, 1787 John Fitch launches a steam-powered rowing boat

1787 French chemist Antoine Lavoisier recognizes the element silicon

October 14, 1787 William Symington demonstrates a paddle steamer

December 3, 1787 James Rumsey tries out a jet-propelled boat

1788 Joseph Louis Lagrange publishes *Mécanique Analytique*